[探索、ゲーム]

[22.2] (1)"agent," "perfect information," "state space," "policy," and "iterative deepening search."

[20.2] (1)"zero-sum games," "deterministic games," "evaluation functions," "thrashing," and "horizon effect."

[25.2] (1) ~~"zero-sum games,"~~ ~~"perfect information,"~~ ~~"policy,"~~ ~~"horizon effect,"~~ and "Monte Carlo tree search."

[22.2] (2)

Describe the algorithm of the Minimax procedure. In addition, explain its completeness, (cost) optimality, time complexity, and space complexity.

[20.2] (2)

Explain what the Minimax procedure is. Draw an example of an AND/OR graph with depth level 2 and branching factor 3, and then use the graph for the explanation.

[20.2] (4)

Explain how to apply the Minimax procedure for a three-player game where each player is adversarial to each other. Make your own example and use it for the explanation.

[25.2] (3)

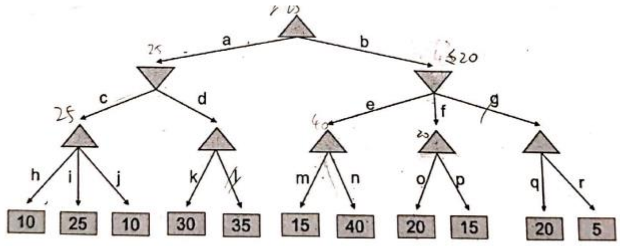
For what kind of games can we apply the minimax procedure to traverse until the terminal nodes? If this traversal is difficult, what can we do instead? Make your own example and use it for the explanation.

[22.2] (4)

Assume that the opponent player is known to play in a random way. Explain a more efficient method than the Minimax procedure.

[25.2] (2)

Write the list of all branches that will be removed if we apply the alpha-beta pruning algorithm to the minimax tree described below. The alpha-beta pruning starts from the root node, and when a node is expanded, it visits the left child first and the right child last. The △ nodes are max nodes, and the ▽ nodes are min nodes. If no branch will be removed, answer "none."



[22.2] (3)

Explain how much the alpha-beta pruning algorithm reduces the time complexity in the most ideal situation. Draw an example, and then use it for the explanation.

[20.2] (3)

Describe the alpha-beta pruning algorithm; in addition, explain to what extent it can improve the efficiency in the best case.

[22.2] (5)

Using keywords such as time complexity, evaluation functions, machine learning, and reinforcement learning, discuss how to realize a strong computer player in various two-player games.

[20.2] (5)

There are reports that computers beat champion-level human players in various games (e.g. chess, shogi, go). With referring to size of search space, number of look-ahead, and evaluation functions, discuss how computers came to beat champion-level human players in various games with different difficulties.

[25.2] (4)

In the future robots may be able to play various games in the real space (for example, table tennis, football, etc.). Discuss to what extent adversarial search techniques like the minimax procedure could be applied in such cases. Also, discuss what other technologies would we need apart from adversarial search techniques.